

A COUNTEREXAMPLE TO THE HIBI–SEYED FAKHARI CONJECTURE ON JOIN–MEET IDEALS

ABSTRACT. Hibi and Seyed Fakhari conjectured that the join–meet ideal of every nonmodular finite lattice is not linearly related. We refute their conjecture with a 12-element lattice. Let L be the bounded rank-three lattice whose atom–coatom incidence graph is the cycle C_{10} . Over $\mathbb{F}_2$, and over every field of characteristic zero,

$$\beta_{1,3}(I_{L,K}) = 149, \quad \beta_{1,j}(I_{L,K}) = 0 \quad (j \geq 4).$$

The vanishings in degrees four and five are certified by exact binary rank computations. The range $j \geq 6$ is not computed: it follows from the regularity bound $\text{reg}(I_{L,K}) = 4$, which is quoted from Hibi and Seyed Fakhari.

1. THE COUNTEREXAMPLE

Let K be a field and L a finite lattice. Set $S_K = K[x_a : a \in L]$. Its join–meet ideal is

$$I_{L,K} = (x_a x_b - x_{a \wedge b} x_{a \vee b} : a, b \in L) \subseteq S_K.$$

Write $\beta_{i,j}(M) = \dim_K \text{Tor}_i^{S_K}(M, K)_j$. An ideal generated in degree two is *linearly related* if $\beta_{1,j} = 0$ for every $j \geq 4$. Hibi and Seyed Fakhari conjectured that $I_{L,K}$ is not linearly related whenever L is nonmodular [1, Conjecture 2.4].

Define

$$L = \{\widehat{0}, \widehat{1}\} \cup \{a_i, b_i : i \in \mathbb{Z}/5\mathbb{Z}\}$$

by taking $a_0, \dots, a_4$ as the atoms and $b_0, \dots, b_4$ as the coatoms, with b_i covering precisely a_i and a_{i+1} . Thus the incidence graph on the ten middle vertices is

$$a_0 - b_0 - a_1 - b_1 - a_2 - b_2 - a_3 - b_3 - a_4 - b_4 - a_0.$$

Indices are read modulo 5.

Theorem 1. *The lattice L is nonmodular. If $K = \mathbb{F}_2$, or if K has characteristic zero, then*

$$\beta_{1,3}(I_{L,K}) = 149, \quad \beta_{1,j}(I_{L,K}) = 0 \quad \text{for every } j \geq 4.$$

Consequently, Conjecture 2.4 of Hibi and Seyed Fakhari is false.

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Proof. Two distinct atoms have a common coatom exactly when they are consecutive, in which case that coatom is unique; otherwise their join is $\widehat{1}$. The dual statement holds for coatoms, and the mixed meets and joins are immediate. Hence the displayed incidence relations define a lattice.

The lattice is not modular. Indeed, $a_0 \leq b_0$, while $a_2 \wedge b_0 = \widehat{0}$ and $a_0 \vee a_2 = \widehat{1}$. Therefore

$$a_0 \vee (a_2 \wedge b_0) = a_0 \neq b_0 = (a_0 \vee a_2) \wedge b_0.$$

Write

$$y = x_{\widehat{0}}, \quad z = x_{\widehat{1}}, \quad A_i = x_{a_i}, \quad B_i = x_{b_i}.$$

If $a \leq b$ then $a \wedge b = a$ and $a \vee b = b$, so a comparable pair contributes the zero binomial. Of the $\binom{12}{2} = 66$ pairs of distinct elements, 31 are comparable and drop out, and the generating set of $I_{L,K}$ is indexed by the remaining 35 incomparable pairs: ten pairs of atoms, ten pairs of coatoms, and fifteen incomparable atom-coatom pairs. The corresponding 35 quadrics consist of

$$A_i A_{i+1} - y B_i, \quad B_i B_{i+1} - A_{i+1} z \quad (i \in \mathbb{Z}/5\mathbb{Z}),$$

one for each of the five pairs of consecutive atoms and each of the five pairs of consecutive coatoms, together with twenty-five binomials $UV - yz$. Here U and V run over the pairs of distinct middle variables whose lattice elements are incomparable with meet $\widehat{0}$ and join $\widehat{1}$, namely the five pairs of nonconsecutive atoms, the five pairs of nonconsecutive coatoms, and the fifteen incomparable atom-coatom pairs. The first monomials of the 35 quadrics are pairwise distinct and none of them occurs as a second monomial, so the 35 quadrics are linearly independent over K ; since they span $(I_{L,K})_2$ and $(I_{L,K})_1 = 0$, they are a K -basis of $(I_{L,K})_2$ and hence a minimal system of generators of $I_{L,K}$.

Let $\mathcal{G}$ be this set of generators, set

$$F = \bigoplus_{g \in \mathcal{G}} S(-2)e_g, \quad \varphi : F \longrightarrow I_{L,K}, \quad e_g \longmapsto g,$$

where $S = S_K$, and write $Z = \ker \varphi$. The degree-two independence gives $Z_2 = 0$.

For $d \geq 2$, define a multigraph Γ_d as follows. For every $g = m_g - n_g \in \mathcal{G}$ and every monomial u of degree $d-2$, add an edge (g, u) joining the degree- d monomials um_g and un_g . The vertices are the monomials incident with at least one edge. After composing φ_d with the inclusion $(I_{L,K})_d \hookrightarrow S_d$ and using the monomial basis of S_d , its matrix is an oriented incidence matrix of Γ_d , with zero rows omitted. Thus, writing E_d, V_d, c_d for the numbers of edges, vertices, and connected components, the graph-incidence rank formula gives

$$\dim_K Z_d = E_d - V_d + c_d.$$

This dimension is independent of K .

Let $\mathfrak{m} = S_+$. Since $(\mathfrak{m}Z)_d = S_1 Z_{d-1}$,

$$\beta_{1,d}(I_{L,K}) = \dim_K Z_d - \text{rank}_K(\mu_d), \quad \mu_d : S_1 \otimes_K Z_{d-1} \longrightarrow Z_d.$$

We now work over $\mathbb{F}_2$. Choose a spanning forest in each Γ_d . Its non-tree edges determine the fundamental-cycle basis of Z_d . In this basis, the coordinate of a cycle at a non-tree edge is its incidence with that edge. Multiplication by a variable sends an edge (g, u) of Γ_{d-1} to the edge $(g, x_r u)$ of Γ_d , so the matrices of μ_4 and μ_5 are obtained exactly by bit-vector arithmetic.

The graph data are

d	E_d	V_d	c_d	$\dim Z_d$
3	420	312	41	149
4	2730	1293	91	1528
5	12740	4276	161	8625

and exact Gaussian elimination over $\mathbb{F}_2$ gives

$$\text{rank}_{\mathbb{F}_2}(\mu_4) = 1528 \quad \text{from a } 1788 \times 1528 \text{ matrix,}$$

$$\text{rank}_{\mathbb{F}_2}(\mu_5) = 8625 \quad \text{from a } 18336 \times 8625 \text{ matrix.}$$

In both displays the matrix is written with its rows indexed by the basis $\{x_r \otimes \zeta\}$ of $S_1 \otimes_K Z_{d-1}$, so that $1788 = 12 \cdot 149$ and $18336 = 12 \cdot 1528$, and with its columns indexed by the fundamental-cycle basis of the target Z_d . In each case the stated rank is therefore the full target rank. Consequently,

$$\beta_{1,3}(I_{L,\mathbb{F}_2}) = 149, \quad \beta_{1,4}(I_{L,\mathbb{F}_2}) = \beta_{1,5}(I_{L,\mathbb{F}_2}) = 0.$$

We now check the hypotheses under which [1, Lemma 2.8] is stated, namely those of [1, Lemma 2.6]. The lattice L is finite and pure of rank three: every maximal chain has the form $\hat{0} < a_i < b_j < \hat{1}$ with $a_i \leq b_j$, and so has length three. Its Hasse diagram is connected, and $|L| = 12 \geq 7$. Finally the graph $G_L := L \setminus \{\hat{0}, \hat{1}\}$ of [1, Section 2] is the atom-coatom incidence graph displayed above, namely the cycle C_{2n} with $2n = 10 \geq 8$, which is connected. All hypotheses hold, so [1, Lemma 2.8] applies and gives $\text{reg}(I_{L,K}) = 4$ over every field. Since $\text{reg}(I) = \max\{j - i : \beta_{i,j}(I) \neq 0\}$, this implies $\beta_{1,j}(I_{L,K}) = 0$ for $j \geq 6$. This completes the proof over $\mathbb{F}_2$.

Finally, orient the graph edges and fundamental cycles over $\mathbb{Z}$, and write $\partial_d : \mathbb{Z}^{E_d} \rightarrow \mathbb{Z}^{V_d}$ for the oriented incidence map of Γ_d . The oriented fundamental cycles form a $\mathbb{Z}$ -basis of the integral cycle lattice $\ker \partial_d$. The image of ∂_d is a subgroup of $\mathbb{Z}^{V_d}$ and hence free, so $\ker \partial_d$ is a direct summand of $\mathbb{Z}^{E_d}$ and $Z_d = K \otimes_{\mathbb{Z}} \ker \partial_d$ for every field K ; the oriented fundamental cycles are thus a K -basis of Z_d for every K , and their reductions modulo 2 are the basis used above. In these integral bases the same construction gives signed integer matrices for μ_4 and μ_5 whose reductions modulo 2 are the two binary matrices above. Full target rank modulo 2 yields a maximal minor with odd determinant, hence a nonzero maximal minor over $\mathbb{Q}$. Both multiplication maps therefore have full target rank over every field of characteristic zero. Together with the field-independent graph dimensions and the regularity bound, this proves the characteristic-zero statement. $\square$

2. EXACT VERIFICATION

Everything above except the two binary ranks can be checked by hand from the cover relations of L , and those two ranks are recomputable from the data printed here. One stores the 20 cover relations $\hat{0} < a_i$, $a_i < b_i$, $a_{i+1} < b_i$, $b_i < \hat{1}$, takes the transitive closure and reads off the 144 meets and joins; reduces $x_a x_b - x_{a \wedge b} x_{a \vee b}$ over the $\binom{12}{2} = 66$ pairs of distinct elements, of which 31 vanish and 35 give the quadrics $\mathcal{G}$ above; forms Γ_d for $d = 3, 4, 5$, obtaining E_d , V_d , c_d and the fundamental-cycle basis by union-find, so that $\dim Z_d = E_d - V_d + c_d$; and for $d = 4, 5$ writes the 12-dim Z_{d-1} products $x_r \zeta$ in that basis and reduces the resulting bit-vector matrix by Gaussian elimination over $\mathbb{F}_2$, which returns the ranks 1528 and 8625 of the two matrices displayed above. All arithmetic is exact, on integers and bit vectors; there is no randomisation, floating-point computation, external solver or tolerance parameter, and no step depends on the order of enumeration. The computation was re-run independently, on a separate machine and by a program written against this manuscript, and it agreed: all 64 of its checks passed, covering the lattice, its purity and rank, its nonmodularity (40 failures of the modular law over all triples $x \leq z$, the witness printed above among them), the 35 generators, every entry of the table, each $\dim Z_d$ recomputed as $E_d - \text{rank}(\partial_d)$ and certified field-independent by exact rank bounds, and every row of μ_4 and μ_5 checked against the cycle space of Γ_d over $\mathbb{Z}$ and over $\mathbb{F}_2$; it also goes one degree beyond this note, obtaining $\dim Z_6 = 35762$ and $\beta_{1,6}(I_{L,K}) = 0$ directly. It does not reach $\beta_{1,j}(I_{L,K}) = 0$ for $j \geq 7$, which no finite computation can settle; the range $j \geq 6$ of Theorem 1 rests instead on $\text{reg}(I_{L,K}) = 4$, quoted from [1, Lemma 2.8] and not recomputed at all. An independent re-run that agrees is evidence about the computation, not a second proof. That program and its transcript accompany this note.

3. STATUS AND SCOPE

Theorem 1 refutes Conjecture 2.4 of [1]: the lattice L is nonmodular and its join-meet ideal is linearly related. Nothing here bears on the other results of [1], and [1, Lemma 2.8], which the proof uses, is taken as correct.

Three limits should be stated. First, $\beta_{1,j}(I_{L,K}) = 0$ is obtained by computation only for $j = 4, 5$; for $j \geq 6$ it is deduced from the quoted regularity bound. Second, the conclusion is established for $K = \mathbb{F}_2$ and for $\text{char } K = 0$. The dimensions $\dim_K Z_d$ are field-independent, but full target rank of μ_4 and μ_5 over a field of characteristic $p \geq 3$ is not claimed, and neither is the value of $\beta_{1,j}(I_{L,K})$ there. Third, no minimality is claimed for L : whether some nonmodular lattice with fewer than twelve elements has a linearly related join-meet ideal is not addressed here.

We are aware of no earlier counterexample. A search of the literature on join-meet ideals and on the linearly related property returned no work that constructs such a lattice or that otherwise settles Conjecture 2.4. That

search cannot be exhaustive, and the result is recorded here as refuting the conjecture rather than as closing the question of priority.

REFERENCES

- [1] T. Hibi and S. A. Seyed Fakhari, *Regularity and depth of binomial ideals arising from combinatorics*, arXiv:2607.18679v1 (2026).